

February 9, 2026

Let's begin by practicing more graphing

e.g. (section 2.1 example 12) Hyperbolic paraboloid

$$g: \mathbb{R}^2 \rightarrow \mathbb{R} \quad g(x, y) = y^2 - x^2 \quad (z = g(x, y) \text{ graph})$$

level curves ($z = c$)

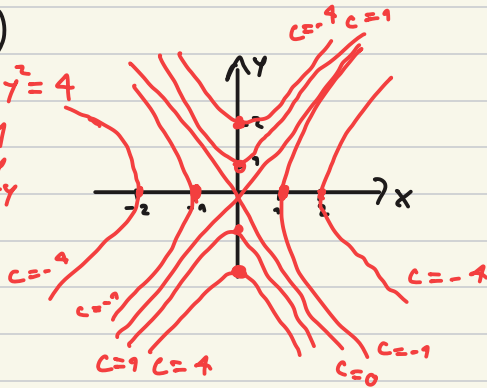
$$c = -4 \leadsto -4 = y^2 - x^2, \quad x^2 - y^2 = 4$$

$$c = -1 \leadsto x^2 - y^2 = 1$$

$$c = 0 \leadsto x^2 = y^2, \quad x = y$$

$$c = 1 \leadsto y^2 - x^2 = 1$$

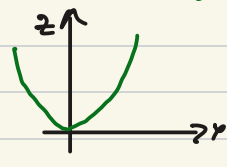
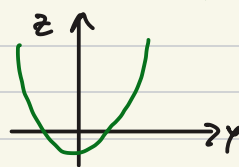
$$c = 4 \leadsto y^2 - x^2 = 4$$



sections parallel to y - z plane ($x = c$)

$$x = -1 \leadsto z = y^2 - 1$$

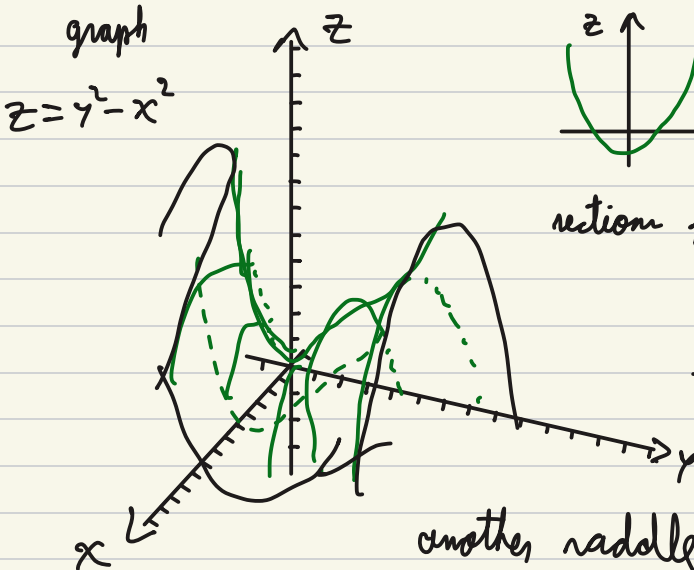
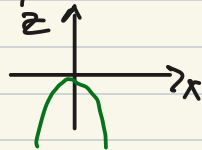
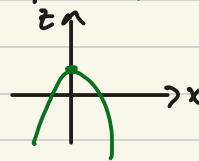
$$x = 0 \leadsto z = y^2 - 0$$



sections parallel to x - z plane ($y = c$)

$$y = -1 \leadsto z = 1 - x^2$$

$$y = 0 \leadsto z = -x^2$$



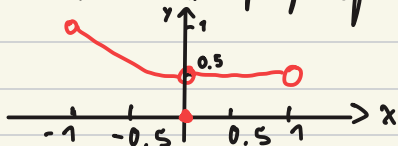
another saddle surface

2.2 Limits

Remember from Calc I and II that limits were crucial to define continuity ($\lim_{x \rightarrow a} f(x) = f(a)$), derivatives ($\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$) and Riemann integrals ($\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{b-a}{n} f(x_i)$). We need them just as much now:

Intuitive idea of limit: For a function $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$, the equation $\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = \vec{L}$ means the image of the function ($f(\vec{x})$) approaches the value $\vec{L} \in \mathbb{R}^n$ (in the codomain) as the independent variable $\vec{x} \in X$ (in the domain) approaches $\vec{a} \in \mathbb{R}^n$ (without ever actually reaching $\vec{a}$).

e.g. This is the graph of some $f: (-1, 1) \subseteq \mathbb{R} \rightarrow \mathbb{R}$



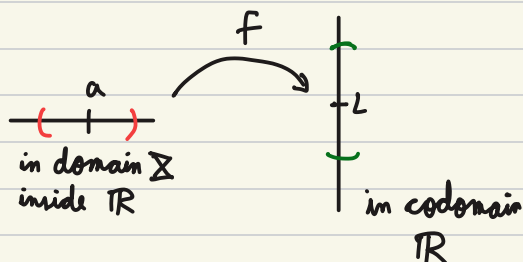
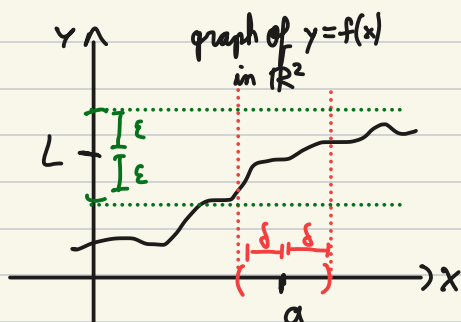
$$\lim_{x \rightarrow 0} f(x) = 0.5 \quad \lim_{x \rightarrow -1} f(x) = 1$$
$$\lim_{x \rightarrow 0.5} f(x) = 0.5$$

definition of limit: Let $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a function, then we say "the limit of f as $\vec{x}$ approaches $\vec{a}$ exists and equals $\vec{L}$ " and express this as $\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = \vec{L}$, if:

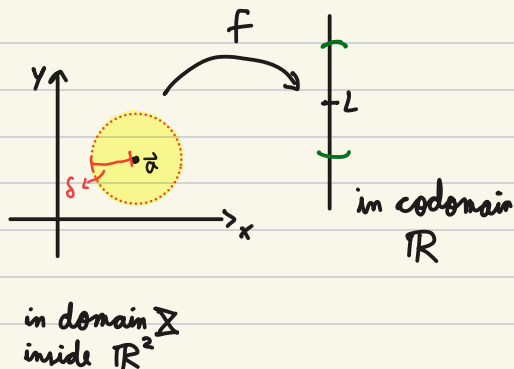
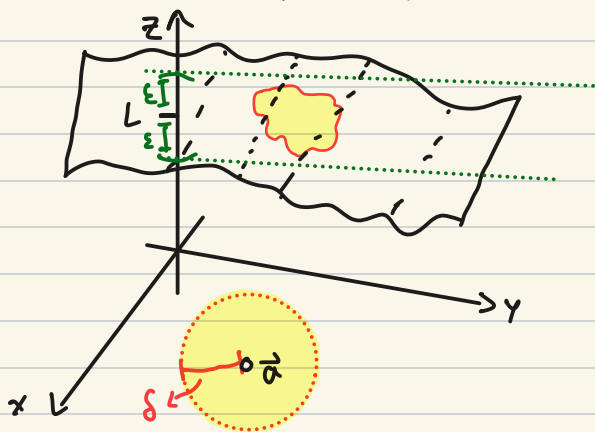
given any $\varepsilon > 0$, you can find a $\delta > 0$ (which depends on ε) such that if $\vec{x} \in X$ and $0 < \|\vec{x} - \vec{a}\| < \delta$, then $\|f(\vec{x}) - \vec{L}\| < \varepsilon$.

How do we visualize limits:

• $f: X \subseteq \mathbb{R} \rightarrow \mathbb{R}$



• $f: X \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$



Theorem: if a limit exists, then it is unique.

e.g. if for $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ $\lim_{\vec{x} \rightarrow \vec{a}} = \vec{L}$ and $\lim_{\vec{x} \rightarrow \vec{a}} = \vec{M}$
then $\vec{L} = \vec{M}$

e.g. Show $\lim_{(x,y) \rightarrow (1,2)} 3x + y = 5$

Fix an arbitrary $\varepsilon > 0$, we want to find $\delta > 0$ such that $0 < \|(x,y) - (1,2)\| < \delta$ implies $\|3x + y - 5\| < \varepsilon$.

Let say we have some $\delta > 0$ and $0 < \|(x,y) - (1,2)\| < \delta$, can we bound $\|f(x,y) - 5\|$?

$$\|(x,y) - (1,2)\| = \sqrt{(x-1)^2 + (y-2)^2} < \delta$$

this implies $|x-1| < \delta$ and $|y-2| < \delta$

$$\text{Hence } |3x-3| < 3\delta \text{ and } |y-2| < \delta,$$

$$\text{so } |3x + y - 5| = |(3x-3) + (y-2)| \leq |3x-3| + |y-2| < 4\delta$$

Now we know $0 < \|(x,y) - (1,2)\| < \delta$ implies $|3x + y - 5| < 4\delta$.

Hence we pick $\delta = \varepsilon/4$ ✓

e.g. Show that $\lim_{(x,y) \rightarrow (a,b)} x = a$ and $\lim_{(x,y) \rightarrow (a,b)} y = b$

e.g. show that $\lim_{(x,y) \rightarrow (a,b)} x = a$ and $\lim_{(x,y) \rightarrow (a,b)} y = b$

Fix $\varepsilon > 0$. Suppose $\|(x,y) - (a,b)\| < \delta$ for some $\delta > 0$,
then $\sqrt{(x-a)^2 + (y-b)^2} < \delta$ and $|x-a| < \delta$, $|y-b| < \delta$

We can pick $\varepsilon = \delta$, so $0 < \|(x,y) - (a,b)\| < \delta$
implies $|x-a| < \varepsilon = \delta$ (this gives $\lim_{(x,y) \rightarrow (a,b)} x = a$)
and $|y-b| < \delta = \varepsilon$ (hence $\lim_{(x,y) \rightarrow (a,b)} y = b$)

Limit laws

Suppose: $\bullet F: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ and $\lim_{\vec{x} \rightarrow \vec{a}} F(\vec{x}) = \vec{L}$

$\bullet G: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ and $\lim_{\vec{x} \rightarrow \vec{a}} G(\vec{x}) = \vec{M}$

$\bullet f: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ and $\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = c$

$\bullet g: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ and $\lim_{\vec{x} \rightarrow \vec{a}} g(\vec{x}) = d$

then:

$$1) \lim_{\vec{x} \rightarrow \vec{a}} F(\vec{x}) + G(\vec{x}) = \lim_{\vec{x} \rightarrow \vec{a}} F(\vec{x}) + \lim_{\vec{x} \rightarrow \vec{a}} G(\vec{x}) = \vec{L} + \vec{M}$$

$$2) \lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) F(\vec{x}) = \left(\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) \right) \left(\lim_{\vec{x} \rightarrow \vec{a}} F(\vec{x}) \right) = c \vec{L}$$

$$3) \text{ if } h \in \mathbb{R}, \text{ then } \lim_{\vec{x} \rightarrow \vec{a}} h F(\vec{x}) = h \lim_{\vec{x} \rightarrow \vec{a}} F(\vec{x}) = h \vec{L}$$

$$4) \lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) g(\vec{x}) = cd$$

$$5) \lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) / g(\vec{x}) = \frac{c}{d} \text{ or long as } d \neq 0 \text{ and } g(\vec{x}) \neq 0$$